Notes: Kang, Rouwenhorst, and Tang (JF, 2020)

A Tale of Two Premiums: The Role of Hedgers and Speculators in Commodity Futures Markets

Contents

1 Big picture

Question. How do the positions and position changes of commercial hedgers and noncommercial speculators affect expected returns in commodity futures markets?

Setting. The paper studies 26 commodity futures markets from January 1994 to December 2017, using weekly Commodity Futures Trading Commission Commitment of Traders data and weekly futures returns.

Core challenge. The traditional normal-backwardation view treats commercial traders as hedgers who are usually net short and who pay speculators a risk premium for price insurance. But the observed net positions of commercials also move because commercials take the other side of short-term trading pressure from noncommercials. The same measured object, "hedging pressure," therefore mixes at least two motives.

Main conceptual contribution. The paper separates two premiums embedded in commodity futures returns:

- a **hedging premium**, paid by commercials to noncommercials for bearing slow-moving price-insurance demand;
- a **liquidity provision premium**, paid by noncommercials to commercials when noncommercials demand short-term immediacy.

Main empirical design. The authors use Fama–MacBeth cross-sectional regressions and portfolio sorts. They ask whether commodities bought or sold by a trader group in week t have predictable returns in weeks $t+1$ and $t+2$. If a group buys commodities that subsequently outperform, that group is interpreted as receiving compensation for providing liquidity. If a group buys commodities that subsequently underperform, that group is interpreted as demanding liquidity.

Main baseline result. Commercial position changes positively predict short-term futures returns. Commodities bought by commercials subsequently outperform commodities sold by commercials. The sign is reversed for noncommercials: commodities bought by noncommercials subsequently underperform. This implies that commercials provide short-term liquidity, while noncommercials consume it.

Key two-premium result. Raw hedging pressure has weak short-term return-predictive power. But once hedging pressure is smoothed over a longer horizon, the positive Keynesian hedging-pressure premium reappears. In the same regression, short-term commercial net buying predicts returns with the opposite economic interpretation: it captures liquidity provision.

Key risk-environment result. The price of liquidity provision rises when commodity-specific volatility is high, when commercials have recently suffered losses, and when commercial positions have become more imbalanced. The effect is not driven by the VIX, which supports the idea that physical-market participants, not mainly financial intermediaries, provide the relevant liquidity.

Momentum result. Noncommercial trading is partly momentum-related, but most short-term noncommercial position changes are orthogonal to recent returns. Commercials lose to the momentum component, but earn enough from providing liquidity to the nonmomentum component that liquidity provision is economically rational.

Economic takeaway. The apparent weakness of hedging pressure in earlier tests comes from mixing two opposite forces. Slow-moving commercial hedging demand raises expected futures returns, while short-term noncommercial liquidity demand lowers expected returns on the commodities noncommercials buy and raises expected returns on the commodities they sell.

2 Data and measurement (key objects)

2.1 CFTC trader-position data

The main data source is the weekly CFTC Commitment of Traders report. The report gives aggregate long and short positions by trader type. Positions are measured on Tuesday and released after the market close on Friday.

The paper uses the standard COT trader categories:

- **commercials**, generally interpreted as hedgers because they use futures for business-related risk management;
- **noncommercials**, generally interpreted as large speculators such as money managers and hedge funds;
- **nonreportables**, smaller traders whose positions are not reported individually.

Interpretation. The labels are useful but imperfect. Commercials may selectively hedge or speculate, and noncommercials may trade for many motives. The paper's central point is that trader labels alone do not reveal trading motives.

2.2 Commodity futures sample

The sample covers 26 commodities traded on four North American exchanges: NYMEX, NYBOT, CBOT, and CME. The sample period is January 2, 1994 to December 31, 2017.

Futures price data come from Pinnacle. Returns are weekly Tuesday-to-Tuesday returns, matching the timing of the CFTC position measurements. The main return uses the front-month contract, with a contract selection rule designed to stay in the liquid part of the futures curve.

Interpretation. The weekly frequency is important. It is short enough to study liquidity provision after position changes, but long enough to align with publicly available CFTC data.

2.3 Hedging pressure

Let $\mathrm{CS}_{i,t}$ be commercial short positions, $\mathrm{CL}_{i,t}$ commercial long positions, and $\mathrm{OI}_{i,t}$ open interest for commodity i in week t. Hedging pressure is defined as

$$\mathrm{HP}_{i,t} = \frac{\mathrm{CS}_{i,t} - \mathrm{CL}_{i,t}}{\mathrm{OI}_{i,t}} = -\frac{\text{commercial net long position}_{i,t}}{\mathrm{OI}_{i,t}}. \tag{1}$$

Interpretation. A high $\mathrm{HP}_{i,t}$ means commercials are more net short. In the normal-backwardation view, higher commercial net short demand should be associated with a higher risk premium to long futures positions.

2.4 Net trading

For a trader category, let $\mathrm{NL}_{i,t}$ denote that category's net long position. Net trading is

$$Q_{i,t} = \frac{\mathrm{NL}_{i,t} - \mathrm{NL}_{i,t-1}}{\mathrm{OI}_{i,t-1}}. \tag{2}$$

Interpretation. $Q_{i,t} > 0$ means the trader group increased its net long futures position during week t. For commercials, if open interest were constant, commercial net buying would correspond to a decrease in measured hedging pressure.

2.5 Propensity to trade

The propensity to trade measures gross trading activity, not just net position changes. For commercials it is

$$\mathrm{PT}_{i,t} = \frac{|\mathrm{CL}_{i,t} - \mathrm{CL}_{i,t-1}| + |\mathrm{CS}_{i,t} - \mathrm{CS}_{i,t-1}|}{\mathrm{CL}_{i,t-1} + \mathrm{CS}_{i,t-1}}. \tag{3}$$

Interpretation. This is analogous to turnover. It measures how actively a trader group changes its positions, regardless of whether the net long position changes much.

2.6 Weekly futures excess return

The weekly excess return on commodity futures is

$$R_{i,t} = \frac{F_i(t,T) - F_i(t-1,T)}{F_i(t-1,T)}, \tag{4}$$

where $F_i(t,T)$ is the futures price at the end of week t for a contract expiring at T.

Interpretation. Since futures require little initial investment, the percentage price change is the relevant excess return for the long futures position.

2.7 Controls for expected returns

The main predictive regressions include standard commodity return predictors:

- $B_{i,t}$: log futures basis, motivated by the theory of storage and carry;
- $R_{i,t}$: current-week return, capturing short-term momentum or reversal;
- $S_{i,t}\widehat{v}_{i,t}$: a signed idiosyncratic risk proxy, where $S_{i,t} = 1$ when noncommercials are net long and $S_{i,t} = -1$ when they are net short, and $\widehat{v}_{i,t}$ is residual volatility from a rolling regression of futures returns on S&P 500 returns.

Interpretation. The controls ensure that the position-change results are not simply rediscovering basis, volatility, or short-term return effects.

2.8 Disaggregated COT data

For robustness, the authors use Disaggregated Commitment of Traders data. DCOT separates traders into producers/merchants/processors/users, money managers, swap dealers, other reportables, and nonreportables.

Interpretation. The DCOT data help sharpen the economic interpretation. Producers are closer to physical hedgers, while money managers are closer to speculative traders. The paper finds that producers provide liquidity and money managers consume it.

2.9 Descriptive facts

Several facts motivate the empirical analysis:

- average excess returns are positive in most commodity markets;
- average hedging pressure is positive in most markets, so commercials are usually net short;
- hedging pressure is volatile, so commercials are not always net short;
- noncommercials have higher propensity to trade than commercials;
- noncommercial position changes are positively related to contemporaneous and lagged futures returns, while commercial position changes are negatively related to those returns.

Interpretation. The data are broadly consistent with commercials as average hedgers, but the high-frequency behavior looks like commercials are contrarians and noncommercials are momentum-style traders.

3 Models and empirical specifications (all equations + interpretation)

3.1 Trading behavior regressions

The paper first studies how position changes relate to returns. For each trader group, the authors run weekly Fama–MacBeth regressions of net trading on contemporaneous or lagged returns and lagged net trading:

$$Q_{i,t}^{g} = \alpha_t^{g} + \lambda_t^{g} R_{i,t-k} + \rho_t^{g} Q_{i,t-1}^{g} + u_{i,t}^{g}, \qquad k \in \{0, 1\}. \tag{5}$$

Here g indexes the trader group.

Interpretation. A positive λ_t^g means trader group g buys recent winners; a negative coefficient means trader group g buys recent losers. The paper finds that noncommercials behave like momentum traders and commercials behave like contrarians.

3.2 Baseline return-predictability regression

The key liquidity test predicts future returns from current position changes:

$$R_{i,t+j} = b_0 + b_1 Q_{i,t} + b_2 B_{i,t} + b_3 S_{i,t} \widehat{v}_{i,t} + b_4 R_{i,t} + \varepsilon_{i,t+j}, \qquad j = 1, 2. \tag{6}$$

The regression is estimated separately using $Q_{i,t}$ for commercials, noncommercials, and nonreportables.

Interpretation. If $b_1 > 0$ for commercials, commodities bought by commercials subsequently earn higher returns. That means commercials receive compensation for taking the other side of others' trades. If $b_1 < 0$ for noncommercials, commodities bought by noncommercials subsequently earn lower returns. That means noncommercials pay a liquidity cost.

3.3 Portfolio-sorting version of the liquidity test

The portfolio test ranks commodities by commercial net trading $Q_{i,t}$ and forms five portfolios. The high-Q portfolio contains commodities most heavily bought by commercials; the low-Q portfolio contains commodities most heavily sold by commercials.

Interpretation. The spread between high-Q and low-Q portfolios measures the return to a liquidity-provision strategy. The paper finds that high-Q commodities outperform shortly after the sort, especially during the first 20 trading days.

3.4 Smoothed hedging pressure

To separate long-run hedging demand from short-run liquidity provision, the paper constructs a slow-moving hedging-pressure measure. Conceptually,

$$\overline{\mathrm{HP}}_{i,t} \approx \frac{1}{52} \sum_{s=0}^{51} \mathrm{HP}_{i,t-s}. \tag{7}$$

Interpretation. The 52-week moving average filters out weekly fluctuations induced by liquidity provision. The remaining component is a proxy for persistent commercial hedging demand.

3.5 Two-premium predictive regression

The main two-premium regression is

$$R_{i,t+j} = b_0 + b_1 \overline{\mathrm{HP}}_{i,t} + b_2 Q^C_{i,t} + b_3 B_{i,t} + b_4 S_{i,t} \widehat{v}_{i,t} + b_5 R_{i,t} + \varepsilon_{i,t+j}, \qquad j = 1, 2, \tag{8}$$

where $Q^C_{i,t}$ is commercial net trading.

Interpretation. b_1 captures the hedging premium: higher persistent commercial net short demand predicts higher returns to long futures. b_2 captures the liquidity premium: commercial net buying predicts short-term outperformance because commercials are compensated for providing liquidity.

3.6 Risk-environment interactions

To test whether liquidity provision becomes more expensive when risk is high, the paper estimates specifications of the form

$$R_{i,t+1} = b_0 + b_1 \overline{\mathrm{HP}}_{i,t} + b_2 Q^C_{i,t} + b_3 Q^C_{i,t} D_{i,t} + \text{controls} + \varepsilon_{i,t+1}. \tag{9}$$

The dummy $D_{i,t}$ can indicate high VIX, high commodity-specific implied volatility, recent losses for commercials, or recent order imbalance.

Interpretation. A positive b_3 means the return impact of commercial net buying is stronger when liquidity is harder to supply. This is what the paper finds for commodity-specific volatility, capital-loss episodes, and order imbalance.

3.7 Momentum and nonmomentum components of noncommercial trading

The paper decomposes noncommercial net trading into a component predicted by the previous week's return and a residual component:

$$Q^N_{i,t} = a + \phi R_{i,t-1} + e_{i,t}. \tag{10}$$

Define

$$Q^{\mathrm{MOM}}_{i,t} = \phi R_{i,t-1}, \qquad Q^{\mathrm{nonMOM}}_{i,t} = a + e_{i,t}. \tag{11}$$

The return-predictability regression becomes

$$R_{i,t+j} = b_0 + b_1 Q^{\mathrm{nonMOM}}_{i,t} + b_2 Q^{\mathrm{MOM}}_{i,t} + b_3 \overline{\mathrm{HP}}_{i,t} + \text{controls} + \varepsilon_{i,t+j}. \tag{12}$$

Interpretation. Q^{MOM} captures trading related to trend-following. Q^{nonMOM} captures the much larger part of noncommercial trading that is unrelated to recent returns. The paper finds that liquidity costs are concentrated in the nonmomentum component, while the momentum component eventually earns positive returns.

3.8 Profit decomposition for commercial traders

Let $np_{i,t}$ be a trader group's net long position scaled by open interest. Let $\overline{np}_{i,t}$ be the slow-moving 52-week average position. The deviation from the smoothed position is

$$Dnp_{i,t} = np_{i,t} - \overline{np}_{i,t}. \tag{13}$$

The paper decomposes this deviation into a momentum-driven component and a liquidity-driven component:

$$Dnp_{i,t} = a + \sum_{j=1}^{52} b_j R_{i,t-j} + \eta_{i,t}. \tag{14}$$

Define

$$Mnp_{i,t} = \sum_{j=1}^{52} b_j R_{i,t-j}, \qquad Lnp_{i,t} = a + \eta_{i,t}. \tag{15}$$

Weekly profit, standardized by dollar open interest, is

$$\begin{aligned} \mathrm{Profit}_{i,t+1} &= np_{i,t} R_{i,t+1} \\ &= \overline{np}_{i,t} R_{i,t+1} + Mnp_{i,t} R_{i,t+1} + Lnp_{i,t} R_{i,t+1}. \end{aligned} \tag{16}$$

Interpretation. The three terms correspond to hedging demand, momentum trading, and liquidity provision. For commercials, the hedging-demand term is typically negative, the momentum term is negative, and the liquidity-provision term is positive and large.

3.9 Standard errors and estimation choices

The core predictive tests use weekly Fama–MacBeth cross-sectional regressions. Reported t-statistics are adjusted using Newey–West standard errors with four lags.

Interpretation. The design exploits weekly cross-sectional variation across commodities while allowing the sequence of weekly slope estimates to be serially correlated.

4 Identification and interpretation (what makes the estimates credible)

4.1 What identifies liquidity provision

The liquidity interpretation comes from the direction of return predictability after trades. If a trader buys and the commodity later underperforms, that trader likely paid a price concession to trade immediately. If a trader buys and the commodity later outperforms, that trader likely supplied liquidity and earned compensation.

Key sign pattern. Commercial net buying predicts positive future returns. Noncommercial net buying predicts negative future returns. This sign pattern is the opposite of a simple story in which noncommercials are merely earning compensation for taking the other side of hedgers.

4.2 Why raw hedging pressure is not enough

Raw hedging pressure combines persistent commercial hedging demand with short-term position movements caused by liquidity provision. These components have opposite effects on expected returns.

Identification logic. Smoothing hedging pressure removes much of the high-frequency liquidity-driven variation. Once the slow component is separated from weekly Q, both premiums become visible.

4.3 Why the result is not just private information

A private-information story would suggest that commercials buy before positive fundamental news and sell before negative fundamental news. But the evidence shows commercials buy recent losers and sell recent winners. Their trades are contrarian and followed by partial reversals.

Interpretation. This is more consistent with liquidity provision than with informed trading. The same logic appears in market microstructure models: liquidity suppliers trade against order flow and earn compensation through subsequent reversals.

4.4 Credibility from DCOT data

The disaggregated data show that producers, merchants, processors, and users provide short-term liquidity, while money managers are the main liquidity consumers. Swap dealers do not appear to be the main liquidity providers.

Interpretation. This supports the paper's claim that liquidity provision is connected to physical commodity market participants, not merely to financial intermediaries classified as commercials.

4.5 Risk-environment evidence

The liquidity premium is stronger when commodity-specific implied volatility is high, when commercials have suffered recent losses, and when their positions have moved in the same direction for several weeks. The VIX interaction is weak.

Interpretation. Liquidity becomes more expensive exactly when it is riskier or costlier for commercials to deviate from their desired exposure.

4.6 What the paper can and cannot prove

The paper cannot directly observe each trader's motive. It therefore interprets motives through position dynamics and return predictability. The central empirical strength is that several independent tests point in the same direction: return reversals, DCOT categories, risk interactions, momentum decomposition, and profit attribution.

Limitation. The evidence is strongest for aggregate trader categories and weekly horizons. It does not identify the exact transaction-level bargaining process or the individual trader initiating each trade.

5 Tables and figures

5.1 Figure 2: Average futures excess returns and average hedging pressure

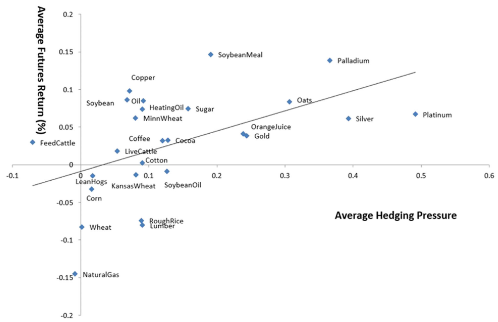


Figure 2. Average futures excess returns and average hedging pressure. The figure provides a scatterplot of the average futures excess return and average hedging pressure for the 26 sample commodities between 1994 and 2017. The cross-sectional regression line has a slope coefficient of 0.27 with a t-statistic of 2.89 and an R^2 of 26%. (Color figure can be viewed at wileyonlinelibrary.com)

Table II

Weekly Position Changes, and Contemporaneous and Lagged Returns

The table reports the average slope coefficients and R^2s of weekly Fama-MacBeth cross-sectional regressions of the net position change (scaled by open interest) $Q_{i,t}$ in week t on an intercept, the contemporaneous futures excess return ($R_{i,t}$) or the lagged return ($R_{i,t-1}$), and the lagged position change ($Q_{i,t-1}$). Separate regressions are run for each of three trader types using CFTC COT classifications: commercials, noncommercials, and others (nonreportables). t-Statistics, reported in parentheses below the coefficients, are adjusted using the Newey-West method with four lags.

	Commercials		Noncommercials		Nonreportables	
$R_{i,t}$	−0.65		0.52		0.13	
	(−37.87)		*(36.07)*		*(20.55)*	
$R_{i,t-1}$		−0.21		0.23		0.02
		(−18.47)		*(22.05)*		*(4.11)*
$Q_{i,t-1}$		0.18		0.16		−0.02
		(19.16)		*(17.38)*		*(−2.54)*
R^2	24.44%	17.79%	20.97%	17.94%	10.19%	12.96%

Read:

- Each point is one commodity.
- The horizontal axis is average hedging pressure.
- The vertical axis is average futures return.
- The fitted relation is positive: commodities with higher average hedging pressure tend to have higher average futures returns.

Economic message. At a long-run average level, the data are consistent with the normal-backwardation idea: when commercials are more persistently net short, long futures investors earn higher average returns.

5.2 Table III: Return predictability following position changes

Read:

- Commercial Q is positive and significant for both week $t + 1$ and week $t + 2$ returns.
- Noncommercial Q is negative and significant.
- The estimated coefficient becomes larger when expected-return controls are included.
- Nonreportable traders have weaker and less stable effects.

Economic message. The sign pattern identifies commercials as liquidity suppliers and noncommercials as liquidity demanders. Noncommercials pay for immediacy through lower subsequent returns on the commodities they buy.

Table III
Return Predictability Following Position Changes: Regression Approach

The table reports the average slope coefficients and R^2s of weekly Fama-MacBeth cross-sectional regressions of the commodity futures excess return in weeks $t+1$ (Panel A) and $t+2$ (Panel B) on an intercept, and the net position change (scaled by open interest) $Q_{i,t}$ in week t, both with and without controls for expected returns. The controls are the log futures basis ($B_{i,t}$), the excess return in week $t(R_{i,t})$, and $S_{i,t}\hat{v}_{i,t}$, where v is the annualized standard deviation of the residuals from a rolling 52-week regression of futures excess returns on SP500 returns and S is an indicator variable equal to one when noncommercials are net long and negative one when they are net short. Separate regressions are run for each of three trader types using CFTC COT classifications: commercials, noncommercials, and nonreportables. t-Statistics, reported in parentheses below the coefficients, are adjusted using the Newey-West method with four lags.

Panel A: Dependent Variable: $R_{i,t+1}$

Coefficient Estimates (×100)	Commercials		Noncommercials		Nonreportables	
$Q_{i,t}$	3.07	4.59	−3.84	−5.24	−1.35	−2.33
	(4.95)	*(6.60)*	*(−5.98)*	*(−7.30)*	*(−0.91)*	*(−1.57)*
$B_{i,t}$		−0.47		−0.47		−0.48
		(−2.71)		*(−2.72)*		*(−2.73)*
$S_{i,t}\hat{v}_{i,t}$		−0.05		−0.02		−0.04
		(−0.41)		*(−0.14)*		*(−0.39)*
$R_{i,t}$		4.51		4.45		2.14
		(4.34)		*(4.33)*		*(2.29)*
R^2	4.82%	25.40%	4.59%	25.28%	4.24%	25.03%

Panel B: Dependent Variable: $R_{i,t+2}$

Coefficient Estimates (×100)	Commercials		Noncommercials		Nonreportables	
$Q_{i,t}$	2.11	3.08	−2.44	−3.49	−2.39	−1.06
	(3.43)	*(4.31)*	*(−3.65)*	*(−4.37)*	*(−1.55)*	*(−0.67)*
$B_{i,t}$		−0.28		−0.28		−0.32
		(−1.63)		*(−1.62)*		*(−1.85)*
$S_{i,t}\hat{v}_{i,t}$		−0.06		−0.06		−0.05
		(−0.51)		*(−0.50)*		*(−0.42)*
$R_{i,t}$		1.98		1.57		−0.13
		(1.83)		*(1.47)*		*(−0.14)*
R^2	4.98%	25.02%	4.85%	24.94%	4.48%	24.77%

Table IV
Weekly Return Predictability Following Position Changes: DCOT Data Set

The table reports the average slope coefficients of weekly Fama-MacBeth cross-sectional regressions of the futures excess return in week $t+1$ (Panel A) and week $t+2$ (Panel B) on an intercept, and the net position change (scaled by open interest) $Q_{i,t}$ in week t, with the same set of control variables as in Table III. Separate regressions are run for each of the trader categories based on the CFTC DCOT classifications: producers/merchants/processors/users, money managers, swap dealers, other reportables, and nonreportables. The sample period is from June 2007 to December 2017. The table reports the time-series average of the slope coefficients and R^2 estimates from weekly cross-sectional regressions. t-Statistics, reported in parentheses below the coefficients, are adjusted using the Newey-West method with four lags.

Panel A: Dependent Variable: $R_{i,t+1}$

Coefficient Estimates (×100)	Producer	Money Manager	Swap Dealer	Other Reportable	Nonreportable
$Q_{i,t}$	8.14	−6.19	−3.32	−0.97	−2.20
	(6.18)	*(−4.39)*	*(−1.20)*	*(−0.40)*	*(−0.77)*
$B_{i,t}$	−0.65	−0.73	−0.73	−0.69	−0.71
	(−2.36)	*(−2.62)*	*(−2.65)*	*(−2.49)*	*(−2.54)*
$S_{i,t}\hat{v}_{i,t}$	−0.06	−0.11	−0.09	−0.09	−0.09
	(−0.30)	*(−0.57)*	*(−0.50)*	*(−0.47)*	*(−0.50)*
$R_{i,t}$	4.51	4.17	1.13	1.36	1.63
	(2.98)	*(2.61)*	*(0.82)*	*(0.97)*	*(1.17)*
R^2	25.39%	25.40%	24.21%	24.45%	24.45%

Panel B: Dependent Variable: $R_{i,t+2}$

Coefficient Estimates (×100)	Producer	Money Manager	Swap Dealer	Other Reportable	Nonreportable
$Q_{i,t}$	4.93	−4.89	−3.55	4.18	−0.11
	(3.77)	*(−3.41)*	*(−1.33)*	*(1.65)*	*(−0.03)*
$B_{i,t}$	−0.52	−0.50	−0.46	−0.50	−0.50
	(−1.84)	*(−1.82)*	*(−1.69)*	*(−1.79)*	*(−1.79)*
$S_{i,t}\hat{v}_{i,t}$	−0.13	−0.12	−0.14	−0.16	−0.12
	(−0.66)	*(−0.63)*	*(−0.77)*	*(−0.82)*	*(−0.62)*
$R_{i,t}$	2.03	1.91	−0.68	0.02	−0.98
	(1.36)	*(1.22)*	*(−0.51)*	*(0.01)*	*(−0.74)*
R^2	24.41%	24.20%	23.35%	23.98%	23.82%

5.3 Table V: Portfolio sorting on commercial position changes

Table V
Return Predictability Following Position Changes: Portfolio Sorting Approach

On Tuesday of each week, commodities are ranked based on the change in the net position of commercial traders, Q. We sort commodities into five "quintile" portfolios containing five, five, six, five, and five commodities each. The table reports the average futures excess returns (Panel A) and average position changes (Panel B) of the commercial traders (normalized by the open interest on the day of ranking) on the quintile portfolios during the 10 trading days prior to ranking and the 40 trading days following the ranking. Because the CFTC measures positions on Tuesdays but publishes the positions after the market close on Friday, we separately calculate the postranking excess returns for days 1 to 4 and days 5 to 40. The t-statistics for the difference in the means of the top and bottom quintiles reported in parentheses, are adjusted using the Newey-West method with four lags.

Panel A: Average Excess Returns (in %)

	−10 to 0 days	1 to 4 days	5 to 10 days	11 to 20 days	21 to 40 days	1 to 40 days
Portfolio 1 (smallest Q)	3.72	−0.05	−0.09	0.08	0.44	0.37
Portfolio 2	1.59	0.02	0.00	0.05	0.16	0.24
Portfolio 3	0.02	0.07	0.07	0.15	0.14	0.42
Portfolio 4	−1.51	0.15	0.07	0.14	0.15	0.50
Portfolio 5 (largest Q)	−3.19	0.19	0.20	0.27	0.34	1.01
Portfolio 5 to 1	−6.91	0.24	0.29	0.19	−0.10	0.64
(t-statistic)		*(3.65)*	*(3.73)*	*(1.68)*	*(−0.58)*	*(2.64)*

Panel B: Average Position Changes of the Commercial Traders (in %)

	−2 to 0 weeks	1 week	2 week	3 to 4 weeks	5 to 8 weeks	1 to 8 weeks
Portfolio 1 (smallest Q)	−7.72	−1.55	−0.22	0.51	0.82	−0.44
Portfolio 2	−2.38	−0.54	−0.22	0.08	0.15	−0.53
Portfolio 3	0.19	−0.04	−0.03	−0.03	0.00	−0.10
Portfolio 4	2.54	0.48	0.05	−0.43	−0.50	−0.39
Portfolio 5 (largest Q)	7.54	1.40	0.06	−0.86	−1.67	−1.07
Portfolio 5 to 1	15.26	2.95	0.28	−1.37	−2.49	−0.63
(t-statistic)		*(26.86)*	*(2.63)*	*(−7.46)*	*(−8.38)*	*(−1.70)*

Read:

- Portfolio 5 contains commodities most heavily bought by commercials.
- Portfolio 1 contains commodities most heavily sold by commercials.
- The high-minus-low spread is positive over the first 20 trading days.
- The position changes partly reverse after the initial shock.

Economic message. The portfolio evidence gives the same message as the regressions. Commercials buy after price pressure and earn a subsequent premium; their short-run liquidity trades then partially unwind.

5.4 Table VI: Smoothed hedging pressure and position changes

Table VI

Return Predictability, Smoothed Hedging Pressure, and Position Changes

The table reports the average slope coefficients and R^2s of weekly Fama-MacBeth cross-sectional regressions of the futures excess return ($R_{i,t+j}$) in week $t+j$ ($j=1,2$) on an intercept, lagged hedging pressure $HP_{i,t}$, lagged smoothed hedging pressure $\overline{HP}_{i,t}$, lagged net position changes of commercials $Q_{i,t}$, and the same control variables as in Table III. t-Statistics, reported in parentheses below the coefficients, are adjusted using the Newey-West method with four lags.

Coefficient Estimates (×100)	Dependent Variable: $R_{i,t+1}$			Dependent Variable: $R_{i,t+2}$		
$HP_{i,t}$	−0.09			0.14		
	(−0.61)			(0.96)		
$\overline{HP}_{i,t}$		0.52	0.47		0.53	0.45
		(3.50)	(2.90)		(3.67)	(2.96)
$Q_{i,t}$			4.75			2.45
			(6.39)			(3.72)
Controls	yes	yes	yes	yes	yes	yes
R^2	25.84%	25.68%	29.78%	25.41%	25.20%	29.46%

Read:

- Raw one-week hedging pressure is not significant.
- Smoothed hedging pressure is positive and significant.
- Commercial net trading Q remains positive and significant when included together with smoothed hedging pressure.
- The two coefficients capture different components of the risk premium.

Economic message. This is the paper's central two-premium result. Slow-moving commercial hedging pressure and short-term commercial liquidity provision both matter, but they reflect different economic forces.

5.5 Table IX: Factors affecting liquidity provision

Table IX

Factors Affecting Liquidity Provision

In Panel A, we measure the incremental return impact of a position change by defining dummy variables $D(VIX)_t$ and $D(ComVol)_{i,t}$ that take the value of one when the VIX is above its full-sample median or the implied volatility for an individual commodity is above its sample median in that week. In Panel B, we define a dummy variable $D(CLoss)_{i,t}$ that is equal to one when the losses (measured over the prior four weeks) of commercial positions in commodity i are above the sample median. $D(OIB)_{i,t}$ is equal to one when the position changes of commercial traders for commodity i were in the same direction (buying or selling) in the prior four weeks from $t-3$ to t. These interactive dummies are part of a predictive panel regression that includes commodity fixed effects: $R_{i,t+1} = b_0 + b_1\overline{HP}_{i,t} + b_2 Q_{i,t} + b_3 Q_{i,t} D(\cdot)_t + controls + \varepsilon_{i,t+1}$. Variables are defined as in Table VII. t-Statistics, reported in parentheses below the coefficients, are adjusted by the Newey-West method with four lags.

Panel A: Liquidity Provision Conditional on VIX and Commodity Volatility

Coefficient Estimates ($\times$100)	Dependent Variable: $R_{i,t+1}$			Dependent Variable: $R_{i,t+2}$		
$Q_{i,t}$	4.08	2.72	3.22	2.41	1.76	1.73
	(5.76)	*(4.87)*	*(4.41)*	*(3.39)*	*(3.09)*	*(2.41)*
$Q_{i,t} \times D(VIX)_t$	−0.73		−0.98	0.26		0.05
	(−0.82)		*(−1.10)*	*(0.29)*		*(0.06)*
$Q_{i,t} \times D(ComVol)_{i,t}$		2.26	2.37		1.88	1.87
		(2.43)	*(2.55)*		*(2.05)*	*(2.04)*
$\overline{HP}_{i,t}$	0.44	0.42	0.42	0.41	0.40	0.40
	(2.28)	*(2.20)*	*(2.19)*	*(2.17)*	*(2.10)*	*(2.10)*
Controls	yes	yes	yes	yes	yes	yes
R^2	0.29%	0.34%	0.34%	0.23%	0.24%	0.24%

Panel B: Capital Constraint and Order Imbalance Effects

Coefficient Estimates ($\times$100)	Dependent Variable: $R_{i,t+1}$		Dependent Variable: $R_{i,t+2}$	
$Q_{i,t}$	2.42	2.83	2.12	1.90
	(4.22)	*(5.72)*	*(3.45)*	*(3.56)*
$Q_{i,t} \times D(CLoss)_{i,t}$	2.17		1.01	
	(2.49)		*(1.16)*	
$Q_{i,t} \times D(OIB)_{i,t}$		2.49		3.35
		(2.29)		*(3.10)*
$\overline{HP}_{i,t}$	0.39	0.33	0.35	0.29
	(2.20)	*(1.85)*	*(1.96)*	*(1.64)*
Controls	yes	yes	yes	yes
R^2	0.31%	0.32%	0.22%	0.26%

Figure 1: Risk environment, capital constraints, and order imbalance.

Read:

- Interacting Q with high VIX does not materially increase the liquidity premium.
- Interacting Q with high commodity-specific volatility does increase the liquidity premium.
- Recent commercial losses also raise the price impact of position changes.
- Order imbalance raises the return impact as well.

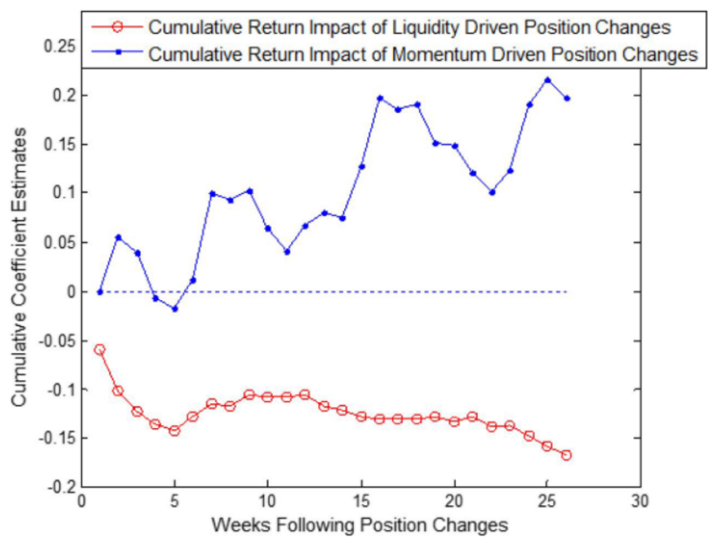
Economic message. Liquidity is more expensive when it is riskier or more balance-sheet intensive for commercials to provide it. The commodity-specific volatility result supports the view that physical-market traders are central liquidity suppliers.

5.6 Table X and Figure 3: Momentum versus nonmomentum position changes

Table X
Returns Following Momentum-Driven versus Nonmomentum Position Changes

We decompose noncommercials' trading measure Q into two components by running the panel regression $Q_{i,t} = a + \varphi \times R_{i,t-1} + e_{i,t}$. We define $Q_{MOM,i,t} = \varphi \times R_{i,t-1}$ as the component of noncommercials' position changes that can predicted by past returns ($R_{i,t-1}$), and $Q_{nonMOM,\ i,t} = a + e_{i,t}$ as the component unrelated to past returns. Next, we use these two components of position changes to predict next-week futures returns. The table reports the average slope coefficients and R^2s of weekly Fama-MacBeth cross-sectional regressions of the futures excess return ($R_{i,t+j}$) in week $t+j$ ($j = 1,2$) on an intercept, the two components of position changes, smoothed lagged hedging pressure $\overline{HP}_{i,t}$, and the same control variables as in Table III. t-Statistics, reported in parentheses below the coefficients, are adjusted using the Newey-West method with four lags.

Coefficient Estimates (×100)	Dependent Variable: $R_{i,t+1}$	Dependent Variable: $R_{i,t+2}$
$Q_{nonMOM,i,t}$	−6.00	−4.13
	(*−7.08*)	(*−4.94*)
$Q_{MOM,i,t}$	−0.11	5.64
	(*−0.03*)	(*1.54*)
$\overline{HP}_{i,t}$	0.44	0.39
	(*2.77*)	(*2.55*)
Controls	yes	yes
R^2	36.05%	35.83%



Read:

- The nonmomentum component of noncommercial trading strongly predicts negative subsequent returns.
- The momentum component is not negative in the same way and becomes positive over longer horizons.
- Figure 3 shows liquidity-driven trades have negative cumulative impact for noncommercials, while momentum-driven trades eventually perform well.

Economic message. Commercials lose to true momentum trading but earn from providing liquidity to the much larger nonmomentum component of short-term noncommercial trading.

5.7 Table XI: Profit attribution of commercial traders

Table XI

Profit Attribution of Commercial Traders

In this table, we decompose the profits and losses of commercial traders into three components: hedging demand, momentum trading, and liquidity provision. We first calculate "Hedging Demand" as the 52-week moving average of past net positions. The deviation between the actual position and the smoothed position is decomposed into a component that can be predicted based on past futures returns and a residual. The former is labeled "Momentum Trading," and the latter is labeled "Liquidity Provision." We multiply the position components described above by the next-week futures return to compute the corresponding profit components. We then take their time-series averages and divide them by the average absolute value of the commercial net positions to obtain a *percentage* annual profit measure from the perspective of the commercial traders. The column "Commercials" provides the average annual percentage profit to commercials averaged across all sample commodities. The next two columns provide a breakout of the offsetting gains and losses to the other traders (i.e., Noncommercials and Nonreportables). In Panel A, we decompose the profit or loss based on the entire available COT sample period from Janurary 2, 1994 to December 31, 2017. In Panel B, we exclude the financial crisis period, from September 15, 2008 (the collapse of Lehman Brothers) to June 30, 2009 (the economic bottom date according to the National Bureau of Economic Research (NBER)), from our profit decomposition analysis.

Panel A: Profit Decomposition Based on the Full-Sample COT Data			
	Commercials	Noncommercials	Nonreportables
Hedging Demand	−4.71%	3.70%	1.01%
Momentum Trading	−1.86%	1.71%	0.15%
Liquidity Provision	5.27%	−3.63%	−1.65%
Total Profit/Loss	−1.30%	1.79%	−0.49%
Panel B: Profit Decomposition Excluding the Financial Crisis			
	Commercials	Noncommercials	Nonreportables
Hedging Demand	−5.09%	4.18%	0.91%
Momentum Trading	−0.50%	0.58%	−0.08%
Liquidity Provision	4.63%	−3.14%	−1.50%
Total Profit/Loss	−0.96%	1.62%	−0.66%

Read:

- Commercials lose on the hedging-demand component.
- Commercials lose on the momentum-trading component.
- Commercials earn a large positive liquidity-provision component.
- Excluding the financial crisis gives a similar pattern.

Economic message. The liquidity premium recovers a large part of what commercials pay for price insurance. This also explains why noncommercial speculative profits can look low: the hedging premium they earn is offset by liquidity costs.

5.8 Robustness tables

Table VII. DCOT data confirm the two-premium result using producer positions instead of aggregated commercial positions.

Table VIII. Two-way sorts on smoothed hedging pressure and commercial net trading show that high smoothed hedging pressure predicts persistent returns, while high commercial net buying predicts short-run returns.

Table XII. Time-series regressions show that a commodity's own position changes predict its own future returns. The cross-sectional strategy's profitability is driven mainly by the own-commodity autocovariance component.

Table XIII. The main return-predictability result is not driven only by minor commodities. It

Table VII

Hedging Pressure and Liquidity Provision: DCOT Data

The table examines the impact of hedging pressure and weekly position changes for risk premiums comparing two different measures of hedging pressure. The first is based on net commercial positions from the Commitment of Traders (COT) reports, the second is based on the net positions of traders in the producers/merchants/processors/users category as reported in the Disaggregate Commitment of Traders (DCOT) reports. The sample period is from June 2007 to December 2017. The table reports the time-series average of slope coefficients and R^2s of weekly Fama-MacBeth cross-sectional regressions of the futures excess return in week $t+1$ (Panel A) and week $t+2$ (Panel B) on an intercept, lagged smoothed hedging pressure ($\overline{HP}_{i,t}$), and lagged net position changes ($Q_{i,t}$) of the producers/merchants/processors/users category, with and without the control variables used in Table III. t-Statistics, reported in parentheses below the coefficients, are adjusted using the Newey-West method with four lags.

Panel A: Dependent Variable: $R_{i,t+1}$				
Coefficient Estimates (×100)	DCOT Data without Controls	DCOT Data with Controls	COT Data without Controls	COT Data with Controls
$\overline{HP}_{i,t}$	0.64	0.48	0.57	0.50
	(2.55)	*(1.73)*	*(2.43)*	*(1.86)*
$Q_{i,t}$	5.16	8.07	4.58	6.68
	(4.21)	*(5.89)*	*(3.59)*	*(4.90)*
Controls	no	yes	no	yes
R^2	10.23%	29.67%	10.89%	29.68%
Panel B: Dependent Variable: $R_{i,t+2}$				
Coefficient Estimates (×100)	DCOT Data without Controls	DCOT Data with Controls	COT Data without Controls	COT Data with Controls
$\overline{HP}_{i,t}$	0.71	0.57	0.57	0.38
	(2.75)	*(2.13)*	*(2.62)*	*(1.63)*
$Q_{i,t}$	2.46	4.49	2.24	4.08
	(2.01)	*(3.27)*	*(1.93)*	*(3.14)*
Controls	no	yes	no	yes
R^2	10.70%	28.75%	10.89%	29.12%

Table VIII

Returns and Position Changes of Double-Sorted Portfolios on Smoothed Hedging Pressure and Position Changes

This table studies commodity futures return predictability based on the previous week's smoothed hedging pressure $\overline{HP}$ and commercial traders' lagged position changes, Q. At the end of each Tuesday, we split our 26 sample commodities into two groups of 13 based on their relative ranking on smoothed hedging pressure $\overline{HP}$. Within each group of 13, we then sort commodities based on the commercial traders' Q, assigning six to the Low Q and seven to the High Q cohort. Panel A reports the average futures excess returns on the double-sorted portfolios, and Panel B reports the average position changes of the commercial traders (normalized by the open interest of the ranking day) in the days and weeks subsequent to the sorting. Since the CFTC measures positions on Tuesdays and publishes the positions after the market close on Friday, we separately calculate the average of the postranking returns for days 1 to 4 and days 5 to 40. t-Statistics for the difference in the means of the top and bottom halves, reported in parentheses, are adjusted using the Newey-West method using four lags.

Panel A: Average Excess Returns (in %)				
	Low Q	High Q	H to L Q	t-Statistic
Days 1 to 4				
Low $\overline{HP}$	−0.10	0.10	0.21	*(3.76)*
High $\overline{HP}$	0.05	0.22	0.17	*(2.89)*
H to L $\overline{HP}$	0.16	0.12		
(t-statistic)	*(2.51)*	*(2.10)*		
Days 5 to 20				
Low $\overline{HP}$	−0.11	−0.06	0.05	*(0.35)*
High $\overline{HP}$	0.20	0.69	0.49	*(3.97)*
H to L $\overline{HP}$	0.31	0.76		
(t-statistic)	*(1.77)*	*(4.85)*		
Days 21 to 40				
Low $\overline{HP}$	−0.08	−0.16	−0.08	*(−0.57)*
High $\overline{HP}$	0.63	0.58	−0.05	*(−0.36)*
H to L $\overline{HP}$	0.71	0.74		
(t-statistic)	*(3.68)*	*(3.68)*		
Days 1 to 40				
Low $\overline{HP}$	−0.29	−0.12	0.17	*(0.83)*
High $\overline{HP}$	0.88	1.49	0.61	*(3.16)*
H to L $\overline{HP}$	1.16	1.60		
(t-statistic)	*(3.57)*	*(5.15)*		

Table VIII—*Continued*

Panel B: Average Position Change of the Commercial Traders (in %)				
	Low Q	High Q	H to L Q	t-Statistic
Week 1				
Low $\overline{HP}$	−0.89	0.35	1.24	*(17.78)*
High $\overline{HP}$	−0.86	0.96	1.82	*(19.62)*
H to L $\overline{HP}$	0.03	0.61		
(t-statistic)	*(0.30)*	*(7.76)*		
Weeks 2 to 4				
Low $\overline{HP}$	−0.51	−0.75	−0.24	*(−1.57)*
High $\overline{HP}$	0.74	−0.22	−0.96	*(−5.10)*
H to L $\overline{HP}$	1.25	0.53		
(t-statistic)	*(5.26)*	*(2.62)*		
Weeks 5 to 8				
Low $\overline{HP}$	−0.16	−1.19	−1.03	*(−5.90)*
High $\overline{HP}$	0.90	−0.28	−1.18	*(−4.82)*
H to L $\overline{HP}$	1.06	0.91		
(t-statistic)	*(3.86)*	*(3.30)*		
Weeks 1 to 8				
Low $\overline{HP}$	−1.56	−1.59	−0.03	*(−0.11)*
High $\overline{HP}$	0.78	0.46	−0.32	*(−0.99)*
H to L $\overline{HP}$	2.34	2.05		
(t-statistic)	*(5.53)*	*(5.20)*		

Table XII

Time-Series Return Predictability

In Panel A, we conduct time-series regressions in which, for each commodity, we regress the futures excess return ($R_{i,t+j}$) in week $t+j$ ($j=1,2$) on an intercept, the net position change (scaled by open interest) $Q_{i,t}$ in week t, and the same control variables as in Table III. We then report the cross-sectional mean and median for the coefficient estimates (multiplied by 100) obtained at the commodity level using the time-series regressions described above. t-Statistics for the mean of the coefficient estimates are reported in parentheses. In Panel B, we decompose the cross-sectional liquidity strategy return R^{XSLIQ} and the time-series liquidity strategy return R^{TSLIQ} into their various components, as proposed in Moskowitz, Ooi, and Pedersen (2012). t-Statistics, reported in parentheses below the coefficients, are adjusted using the Newey-West method with four lags.

Panel A: Time-Series Futures Return-Prediction Regressions				
Dependent Variable $= R_{i,t+1}$				
Commercials				
	$Q_{i,t}$	$B_{i,t}$	$S_{i,t}\hat{v}_{i,t}$	$R_{i,t}$
Mean	3.95	−0.29	−0.08	1.98
(t-statistic)	*(4.30)*	*(−1.05)*	*(−0.77)*	*(2.51)*
Median	3.63	−0.09	−0.06	2.52
Noncommercials				
	$Q_{i,t}$	$B_{i,t}$	$S_{i,t}\hat{v}_{i,t}$	$R_{i,t}$
Mean	−4.70	−0.28	−0.06	1.74
(t-statistic)	*(−4.58)*	*(−1.01)*	*(−0.62)*	*(2.41)*
Median	−3.63	−0.12	−0.02	2.39
Dependent Variable $= R_{i,t+2}$				
Commercials				
	$Q_{i,t}$	$B_{i,t}$	$S_{i,t}\hat{v}_{i,t}$	$R_{i,t}$
Mean	1.94	−0.41	−0.06	1.01
(t-statistic)	*(2.37)*	*(−1.46)*	*(−0.64)*	*(0.89)*
Median	1.49	−0.24	−0.02	1.32
Noncommercials				
	$Q_{i,t}$	$B_{i,t}$	$S_{i,t}\hat{v}_{i,t}$	$R_{i,t}$
Mean	−1.82	−0.42	−0.05	0.52
(t-statistic)	*(−1.68)*	*(−1.48)*	*(−0.53)*	*(0.47)*
Median	−1.52	−0.26	−0.02	0.74

Panel B: Return Decompositions			
Cross-sectional strategy weekly return: R^{XSLIQ}			
Auto	Cross	Mean level	Total
0.21%	−0.02%	−0.00%	0.18%
Time-series strategy weekly return: R^{TSLIQ}			
Auto	Mean Squared	Total	
0.21%	−0.00%	0.21%	

Table XIII

Return Predictability: Major and Minor Commodities

In this table, we conduct a Fama-MacBeth regression of commodity futures excess returns ($R_{i,t+j}$) in week $t+j$ ($j=1,2$) on an intercept, the net position change (scaled by open interest) $Q_{i,t}$ in week t, and the same control variables as in Table III, separately for the subsamples of major and minor commodities. We divide the 26 sample commodities into two equal halves according to the average total number of traders or the average open interest, both of which are available from the COT database. We report the time-series average of the weekly cross-sectional regression coefficient estimates and the average R^2s for these two subsamples. t-Statistics, reported in parentheses below the coefficients, are adjusted using the Newey-West method with four lags.

Coefficient Estimates (×100)	Major: Commercials	Major: Noncommercials	Minor: Commercials	Minor: Noncommercials
Major and Minor Commodities Classified by Average Total Number of Traders				
Dependent Variable: $R_{i,t+1}$				
$Q_{i,t}$	4.79	−4.92	5.16	−5.44
	(3.77)	*(−3.54)*	*(5.28)*	*(−5.31)*
Controls	yes	Yes	yes	yes
R^2	43.82%	43.98%	39.51%	39.73%
Dependent Variable: $R_{i,t+2}$				
$Q_{i,t}$	2.02	−2.28	4.68	−5.23
	(1.55)	*(−1.55)*	*(4.61)*	*(−4.85)*
Controls	yes	Yes	yes	yes
R^2	42.93%	42.92%	40.08%	40.01%
Major and Minor Commodities Classified by Average Open Interest				
Dependent Variable: $R_{i,t+1}$				
$Q_{i,t}$	5.18	−4.90	4.10	−5.30
	(4.27)	*(−3.55)*	*(3.95)*	*(−4.88)*
Controls	yes	yes	yes	yes
R^2	44.50%	44.41%	39.65%	39.56%
Dependent Variable: $R_{i,t+2}$				
$Q_{i,t}$	4.39	−4.67	3.08	−4.20
	(2.89)	*(−2.88)*	*(3.02)*	*(−3.60)*
Controls	yes	yes	yes	yes
R^2	44.82%	44.65%	39.25%	39.20%

appears in major and minor commodity subsamples, though magnitudes differ by market-size definition.

6 Short summary

- Commercials are usually net short, and average hedging pressure is positively related to average futures returns.
- But weekly commercial position changes are not only hedging demand. They often reflect liquidity provision to noncommercial traders.
- Noncommercials buy recent winners and sell recent losers; commercials trade in the opposite direction.
- Commercial net buying predicts higher future returns, while noncommercial net buying predicts lower future returns.
- Raw hedging pressure fails to predict well because it mixes slow hedging demand with short-term liquidity-driven position changes.
- Smoothed hedging pressure captures the hedging premium; weekly position changes capture the liquidity premium.
- Liquidity provision is costlier when commodity-specific volatility is high, when commercials face losses, and when order imbalance is large.
- Commercials lose to momentum traders but earn enough from liquidity provision to make the activity economically sensible.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

Commodity futures returns contain two position-related premiums. First, persistent commercial hedging pressure is associated with a traditional hedging premium paid to long futures investors. Second, short-term noncommercial trading demand creates a liquidity premium earned by commercials who accommodate that demand.

7.2 Economic intuition

Commercials want price insurance, so they are often willing to pay noncommercials for bearing commodity risk. But commercials also have flexibility in the timing and intensity of their futures positions. When noncommercials demand immediacy, commercials can temporarily move away from their desired exposure and charge a liquidity premium.

7.3 Takeaway

The paper reconciles two views of commodity futures markets. Commercials are hedgers in the long run, but they are also liquidity providers in the short run. Ignoring this distinction makes hedging pressure look weak; separating the two components reveals both the hedging premium and the liquidity premium.